

CO5_AT3 - HACKATHON & MINI PROJECT

DISASTER PREPAREDNESS AND RESPONSE EDUCATION SYSTEM FOR SCHOOLS AND COLLEGES

COMPLETE SOURCE CODE + DATABASE + OUTPUT SCREENSHOTS

Technology: Python, Flask, HTML, CSS, JavaScript, MySQL and Bootstrap

Submission Type: Single PDF File

Project purpose: interactive disaster learning modules, quizzes, mock emergency scenarios, progress tracking, scores, badges/certificates and teacher/admin monitoring.

1. ALLOTTED PROBLEM AND REQUIREMENTS

Allotted Topic: Disaster Preparedness and Response Education System for Schools and Colleges

Problem Requirement: Develop an educational software platform that teaches students about different disasters, safety procedures and emergency-response methods through learning modules, interactive quizzes, mock scenarios, progress tracking, scores, badges or certificates and teacher/admin dashboards.

Required Software: Python, Flask/Django, HTML, CSS, JavaScript, MySQL and Bootstrap.

This submission implements the Flask option and provides MySQL database configuration. The report contains the source code and sample output screens in one PDF, matching the upload requirement shown in the assignment portal.

2. IMPLEMENTED FEATURES

- Student/Admin login
- Earthquake, flood, fire and cyclone learning modules
- Safety procedures before/during/after disasters
- Interactive multiple-choice quiz with score
- Mock fire-emergency decision scenario
- Progress and points tracking
- Preparedness certificate
- Teacher/Admin monitoring dashboard
- MySQL database schema and configuration

3. PROJECT FOLDER STRUCTURE

```
Disaster_Preparedness_Project/  
|-- app.py  
|-- requirements.txt  
|-- schema.sql  
|-- README.txt  
|-- templates/  
|   |-- base.html  
|   |-- login.html  
|   |-- dashboard.html  
|   |-- modules.html  
|   |-- module_detail.html  
|   |-- quiz.html  
|   |-- scenario.html  
|   |-- admin.html  
|   |-- certificate.html  
|-- static/  
|   |-- css/
```

^-- style.css

4. PYTHON FLASK BACKEND - app.py

```
from flask import Flask, render_template, request, redirect, url_for, session, flash
from flask_sqlalchemy import SQLAlchemy
from werkzeug.security import generate_password_hash, check_password_hash
from datetime import datetime
import os

app = Flask(__name__)
app.secret_key = os.getenv('SECRET_KEY', 'disaster-edu-secret')
# MySQL example: mysql+pymysql://root:password@localhost/disaster_education
app.config['SQLALCHEMY_DATABASE_URI'] = os.getenv('DATABASE_URL', 'sqlite:///disaster_education.db')
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
db = SQLAlchemy(app)

class User(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(100), nullable=False)
    email = db.Column(db.String(120), unique=True, nullable=False)
    password = db.Column(db.String(255), nullable=False)
    role = db.Column(db.String(20), default='student')
    points = db.Column(db.Integer, default=0)

class Progress(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey('user.id'), nullable=False)
    module = db.Column(db.String(50), nullable=False)
    completed = db.Column(db.Boolean, default=False)
    score = db.Column(db.Integer, default=0)
    updated_at = db.Column(db.DateTime, default=datetime.utcnow)

MODULES = [
    {'id': 'earthquake', 'title': 'Earthquake Safety', 'icon': '🌍', 'desc': 'Drop, Cover and Hold On; evacuation and aftershock safety.'},
    {'id': 'flood', 'title': 'Flood Preparedness', 'icon': '🌊', 'desc': 'Warning signs, safe evacuation routes and electrical safety.'},
    {'id': 'fire', 'title': 'Fire Emergency', 'icon': '🔥', 'desc': 'RACE method, extinguisher awareness and safe exit procedure.'},
    {'id': 'cyclone', 'title': 'Cyclone & Severe Weather', 'icon': '🌀', 'desc': 'Alerts, shelter preparation and emergency supplies.'},
]

QUIZ = [
    ('During an earthquake indoors, the safest first action is?', ['Run outside immediately', 'Drop, Cover and Hold On', 'Use the lift', 'Stand near a window'], 1),
    ('During a flood, which action is safest?', ['Walk through moving water', 'Touch wet electrical equipment', 'Move to higher ground', 'Drive through flooded roads'], 2),
    ('If clothes catch fire, what should you do?', ['Run fast', 'Stop, Drop and Roll', 'Hide under a table', 'Use an elevator'], 1),
    ('During a cyclone warning, you should?', ['Stay near windows', 'Ignore official alerts', 'Move to a safe shelter', 'Go to the roof'], 2),
]

@app.before_request
def setup_database():
    db.create_all()
    if not User.query.filter_by(email='student@demo.com').first():
        db.session.add(User(name='Demo Student', email='student@demo.com', password=generate_password_hash('student123'), role='student', points=120))
        db.session.add(User(name='Teacher Admin', email='admin@demo.com', password=generate_password_hash('admin123'), role='admin', points=0))
    db.session.commit()
```

```

@app.route('/')
def home():
    if session.get('user_id'):
        return redirect(url_for('dashboard'))
    return render_template('login.html')

@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        user = User.query.filter_by(email=request.form['email']).first()
        if user and check_password_hash(user.password, request.form['password']):
            session['user_id'], session['name'], session['role'] = user.id, user.name, user.role
            return redirect(url_for('dashboard'))
        flash('Invalid email or password', 'danger')
    return render_template('login.html')

@app.route('/logout')
def logout():
    session.clear(); return redirect(url_for('home'))

def current_user():
    return db.session.get(User, session.get('user_id')) if session.get('user_id') else None

@app.route('/dashboard')
def dashboard():
    user=current_user()
    if not user: return redirect(url_for('login'))
    records=Progress.query.filter_by(user_id=user.id).all()
    done=sum(1 for r in records if r.completed)
    best=max([r.score for r in records] or [0])
    completion=int(done/len(MODULES)*100)
    return render_template('dashboard.html', user=user, modules=MODULES, done=done, best=best,
completion=completion)

@app.route('/modules')
def modules():
    if not current_user(): return redirect(url_for('login'))
    return render_template('modules.html', modules=MODULES)

@app.route('/module/<module_id>')
def module_detail(module_id):
    if not current_user(): return redirect(url_for('login'))
    m=next((x for x in MODULES if x['id']==module_id), None)
    if not m: return redirect(url_for('modules'))
    lessons={
        'earthquake':['Before: secure shelves and know exits.','During: Drop, Cover and Hold On.','After:
evacuate calmly and expect aftershocks.'],
        'flood':['Watch official warnings.','Move to higher ground early.','Never walk or drive through
moving floodwater.'],
        'fire':['Raise alarm and use nearest safe exit.','Crawl low under smoke.','Never use lifts during a
fire.'],
        'cyclone':['Prepare emergency supplies.','Stay indoors away from windows.','Follow official
evacuation/shelter instructions.']}
    return render_template('module_detail.html', m=m, lessons=lessons[module_id])

@app.route('/complete/<module_id>')
def complete(module_id):
    user=current_user()
    if not user: return redirect(url_for('login'))
    r=Progress.query.filter_by(user_id=user.id,module=module_id).first()
    if not r: r=Progress(user_id=user.id,module=module_id,completed=True); db.session.add(r)
    else: r.completed=True
    user.points += 25

```

```

db.session.commit(); flash('Module completed! +25 points', 'success')
return redirect(url_for('dashboard'))

@app.route('/quiz', methods=['GET', 'POST'])
def quiz():
    user=current_user()
    if not user: return redirect(url_for('login'))
    score=None
    if request.method=='POST':
        correct=0
        for i, (_,_, answer) in enumerate(QUIZ):
            if request.form.get(f'q{i}') == str(answer): correct += 1
        score=int(correct/len(QUIZ)*100)
        r=Progress.query.filter_by(user_id=user.id,module='quiz').first()
        if not r: r=Progress(user_id=user.id,module='quiz'); db.session.add(r)
        r.score=max(r.score,score); r.completed=True
        user.points += correct*10
        db.session.commit()
    return render_template('quiz.html', quiz=QUIZ, score=score)

@app.route('/scenario', methods=['GET', 'POST'])
def scenario():
    if not current_user(): return redirect(url_for('login'))
    result=None
    if request.method=='POST':
        choice=request.form.get('choice')
        result = 'Excellent response: raise the alarm, use the safe staircase, and assemble at the
designated point.' if choice=='safe' else 'Unsafe choice. During a fire emergency, never use the lift or
return for belongings.'
    return render_template('scenario.html', result=result)

@app.route('/admin')
def admin():
    user=current_user()
    if not user or user.role!='admin': return redirect(url_for('dashboard'))
    users=User.query.filter_by(role='student').all()
    data=[]
    for u in users:
        recs=Progress.query.filter_by(user_id=u.id).all()
        data.append({'name':u.name,'email':u.email,'points':u.points,'completed':sum(r.completed for r in
recs),'best':max([r.score for r in recs] or [0])})
    return render_template('admin.html', data=data)

@app.route('/certificate')
def certificate():
    user=current_user()
    if not user: return redirect(url_for('login'))
    return render_template('certificate.html', user=user, date=datetime.now().strftime('%d %B %Y'))

if __name__ == '__main__':
    app.run(debug=True)

```

5. BASE TEMPLATE - templates/base.html

```
<!doctype html><html><head><meta charset="utf-8"><meta name="viewport" content="width=device-  
width,initial-scale=1"><title>{% block title %}SafeCampus{% endblock %}</title><link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet"><link  
rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}"></head><body>  
<nav class="navbar navbar-expand-lg navbar-dark bg-primary shadow-sm"><div class="container"><a  
class="navbar-brand fw-bold" href="{{ url_for('dashboard') }}"><img alt="SafeCampus logo" data-bbox="278 155 295 172"/> SafeCampus</a>{% if  
session.get('user_id') %}<div class="ms-auto d-flex gap-2"><a class="btn btn-sm btn-light"  
href="{{ url_for('modules') }}">Modules</a><a class="btn btn-sm btn-warning"  
href="{{ url_for('quiz') }}">Quiz</a><a class="btn btn-sm btn-outline-light" href="{{ url_for('scenario')  
 }}">Mock Scenario</a>{% if session.get('role')=='admin' %}<a class="btn btn-sm btn-dark"  
href="{{ url_for('admin') }}">Admin</a>{% endif %}<a class="btn btn-sm btn-danger"  
href="{{ url_for('logout') }}">Logout</a></div>{% endif %}</div></nav>  
<main class="container py-4">{% with msgs=get_flashed_messages(with_categories=true) %}{% for c,m in msgs  
 %}<div class="alert alert-{{c}}">{{m}}</div>{% endfor %}{% endwith %}{% block content %}{% endblock  
 %}</main></body></html>
```

6. LOGIN PAGE - templates/login.html

```
{% extends 'base.html' %}{% block title %}Login - SafeCampus{% endblock %}{% block content %}<div
class="login-wrap"><div class="row g-0 shadow-lg rounded-4 overflow-hidden bg-white"><div class="col-md-6
hero p-5 text-white"><h1 class="display-5 fw-bold">Disaster Preparedness & Response Education</h1><p
class="lead">Learn. Practice. Prepare. Respond safely.</p><div class="mt-4"><div class="col-md-6 p-5"><h2 class="fw-bold">Student / Admin
Login</h2><p class="text-muted">Access learning modules, quizzes and mock scenarios.</p><form
method="post" action="{{ url_for('login') }}"><label class="form-label">Email</label><input class="form-
control mb-3" name="email" value="student@demo.com" required><label
class="form-label">Password</label><input class="form-control mb-3" type="password" name="password"
value="student123" required><button class="btn btn-primary w-100 py-2">Login
Securely</button></form><small class="text-muted d-block mt-3">Demo student: student@demo.com /
student123<br>Admin: admin@demo.com / admin123</small></div></div></div>{% endblock %}
```


7. STUDENT DASHBOARD - templates/dashboard.html

```
{% extends 'base.html' %}{% block content %}<div class="d-flex justify-content-between align-items-center mb-4"><div><h1 class="fw-bold">Welcome, {{user.name}} 🖐️</h1><p class="text-muted">Your preparedness learning dashboard</p></div><div class="badge bg-warning text-dark fs-5 p-3">★ {{user.points}} Points</div></div><div class="row g-3 mb-4"><div class="col-md-4"><div class="stat"><span>Learning Progress</span><b>{{completion}}%</b></div></div><div class="col-md-4"><div class="stat"><span>Modules Completed</span><b>{{done}} / {{modules|length}}</b></div></div><div class="col-md-4"><div class="stat"><span>Best Quiz Score</span><b>{{best}}%</b></div></div></div><div class="card p-4 border-0 shadow-sm mb-4"><h4 class="fw-bold">Preparedness Progress</h4><div class="progress" style="height:22px"><div class="progress-bar progress-bar-striped" style="width:{{completion}}%">{{completion}}%</div></div></div><h3 class="fw-bold mb-3">Continue Learning</h3><div class="row g-3">{% for m in modules %}<div class="col-md-3"><div class="module-card h-100"><div class="display-4">{{m.icon}}</div><h5>{{m.title}}</h5><p>{{m.desc}}</p><a class="btn btn-outline-primary" href="{{ url_for('module_detail', module_id=m.id) }}">Start Module</a></div></div>{% endfor %}</div><div class="mt-4"><a class="btn btn-success" href="{{url_for('certificate')}}">🏆 View Certificate</a></div>{% endblock %}
```

8. LEARNING MODULES - templates/modules.html

```
{% extends 'base.html' %}{% block content %}<h1 class="fw-bold">Disaster Learning Modules</h1><p
class="text-muted">Choose a module and learn what to do before, during and after an emergency.</p><div
class="row g-4 mt-2">{% for m in modules %}<div class="col-md-6"><div class="module-card h-100"><div
class="d-flex align-items-center gap-3"><span
class="display-4">{{m.icon}}</span><div><h3>{{m.title}}</h3><p class="mb-2">{{m.desc}}</p><a class="btn
btn-primary" href="{{url_for('module_detail',module_id=m.id)}}">Open Lesson</a></div></div></div>{%
endfor %}</div>{% endblock %}
```

9. MODULE DETAIL - templates/module_detail.html

```
{% extends 'base.html' %}{% block content %}<div class="card border-0 shadow-sm p-5"><div
class="display-3">{{m.icon}}</div><h1 class="fw-bold">{{m.title}}</h1><p
class="lead">{{m.desc}}</p><hr><h4>Safety Procedure</h4><ol class="list-group list-group-numbered
mb-4">{% for x in lessons %}<li class="list-group-item py-3">{{x}}</li>{% endfor %}</ol><div class="alert
alert-info"><b>Remember:</b> Stay calm, follow trained staff and official emergency instructions.</div><a
class="btn btn-success btn-lg" href="{{url_for('complete',module_id=m.id)}}">✓ Mark Module
Complete</a></div>{% endblock %}
```

10. INTERACTIVE QUIZ - templates/quiz.html

```
{% extends 'base.html' %}{% block content %}<h1 class="fw-bold">Interactive Preparedness Quiz</h1><p
class="text-muted">Test your emergency-response knowledge.</p>{% if score is not none %}<div class="alert
alert-{{ 'success' if score>=75 else 'warning' }} fs-5">Your Score: <b>{{score}}%</b> – {{ 'Preparedness
badge earned!' if score>=75 else 'Review the modules and try again.' }}</div>{% endif %}<form
method="post">{% for q,opts,ans in quiz %}<div class="card border-0 shadow-sm p-4
mb-3"><h5>{{loop.index}}. {{q}}</h5>{% set qi=loop.index0 %}{% for o in opts %}<label class="quiz-
option"><input type="radio" name="q{{qi}}" value="{{loop.index0}}" required> {{o}}</label>{% endfor
%}</div>{% endfor %}<button class="btn btn-primary btn-lg">Submit Quiz</button></form>{% endblock %}
```

11. MOCK SCENARIO - templates/scenario.html

```
{% extends 'base.html' %}{% block content %}<h1 class="fw-bold">🔥 Mock Emergency Scenario</h1><div
class="scenario-box"><h3>Scenario: Fire alarm during class</h3><p class="lead">Smoke is visible near one
corridor. You are on the second floor. What should you do?</p><form method="post"><button name="choice"
value="unsafe" class="btn btn-outline-danger m-2">Use the lift to leave quickly</button><button
name="choice" value="safe" class="btn btn-success m-2">Raise alarm, use safe staircase &
assemble</button><button name="choice" value="unsafe" class="btn btn-outline-danger m-2">Return to
collect your bag</button></form>{% if result %}<div class="alert alert-info mt-4 fs-5">{{result}}</div>{%
endif %}</div>{% endblock %}
```

12. ADMIN DASHBOARD - templates/admin.html

```
{% extends 'base.html' %}{% block content %}<h1 class="fw-bold">Teacher / Admin Dashboard</h1><p
class="text-muted">Monitor participation, scores and preparedness progress.</p><div class="row g-3
mb-4"><div class="col-md-4"><div class="stat"><span>Total
Students</span><b>{{data|length}}</b></div></div><div class="col-md-4"><div class="stat"><span>Active
Learning</span><b>100%</b></div></div><div class="col-md-4"><div class="stat"><span>Preparedness
Status</span><b>GOOD</b></div></div></div><div class="card border-0 shadow-sm p-4"><table class="table
table-hover"><thead><tr><th>Student</th><th>Email</th><th>Modules</th><th>Best
Score</th><th>Points</th></tr></thead><tbody>{% for x in data
%}<tr><td>{{x.name}}</td><td>{{x.email}}</td><td>{{x.completed}}</td><td>{{x.best}}%</td><td>★
{{x.points}}</td></tr>{% endfor %}</tbody></table></div>{% endblock %}
```

13. CERTIFICATE - templates/certificate.html

```
{% extends 'base.html' %}{% block content %}<div class="certificate text-center"><div
class="display-1">🏆</div><h1 class="display-4 fw-bold">Certificate of Preparedness</h1><p
class="lead">This certificate is presented to</p><h2 class="text-primary fw-bold">{{user.name}}</h2><p
class="lead">for participating in the Disaster Preparedness and Response Education
Program.</p><hr><p>Issued on {{date}}</p><p class="fw-bold">SafeCampus - Schools & Colleges Emergency
Education</p></div>{% endblock %}
```

14. CSS DESIGN - static/css/style.css

```
body{background:#f4f7fb;color:#182230}.login-wrap{max-width:1050px;margin:55px auto}.hero{background:linear-gradient(135deg,#0d6efd,#5b38db);min-height:440px;display:flex;flex-direction:column;justify-content:center}.stat,.module-card{background:#fff;border-radius:16px;padding:24px;box-shadow:0 8px 28px rgba(20,43,80,.08)}.stat{display:flex;justify-content:space-between;align-items:center}.stat b{font-size:28px;color:#0d6efd}.module-card p{min-height:58px;color:#6c757d}.quiz-option{display:block;padding:10px 14px;border:1px solid #dde5ef;border-radius:10px;margin:8px 0;background:#fbfddf}.scenario-box{background:#fff;border-radius:20px;padding:45px;box-shadow:0 8px 30px rgba(0,0,0,.08)}.certificate{background:#fff;border:10px double #e0aa22;padding:70px 50px;margin:20px auto;max-width:900px;box-shadow:0 10px 35px rgba(0,0,0,.1)}
```


15. MYSQL DATABASE - schema.sql

```
CREATE DATABASE disaster_education;
USE disaster_education;
CREATE TABLE users (id INT AUTO_INCREMENT PRIMARY KEY,name VARCHAR(100) NOT NULL,email VARCHAR(120)
UNIQUE NOT NULL,password VARCHAR(255) NOT NULL,role VARCHAR(20) DEFAULT 'student',points INT DEFAULT 0);
CREATE TABLE progress (id INT AUTO_INCREMENT PRIMARY KEY,user_id INT NOT NULL,module VARCHAR(50) NOT
NULL,completed BOOLEAN DEFAULT FALSE,score INT DEFAULT 0,updated_at DATETIME DEFAULT
CURRENT_TIMESTAMP,FOREIGN KEY (user_id) REFERENCES users(id));
```

16. REQUIREMENTS - requirements.txt

```
Flask==3.0.3  
Flask-SQLAlchemy==3.1.1  
PyMySQL==1.1.1  
Werkzeug==3.0.3
```

17. HOW TO RUN THE PROJECT

1. Install Python 3.10 or later and MySQL Server.
2. Open Command Prompt inside the project folder.
3. Install packages: `pip install -r requirements.txt`
4. Open MySQL and execute `schema.sql` to create the `disaster_education` database.
5. Set the database URL in Windows CMD: set
`DATABASE_URL=mysql+pymysql://root:YOUR_PASSWORD@localhost/disaster_education`
6. Run the application: `python app.py`
7. Open `http://127.0.0.1:5000` in the browser.
8. Use `student@demo.com / student123` for student output. Use `admin@demo.com / admin123` for admin output.

Note: The source code contains a local SQLite fallback only to make development/testing easier if DATABASE_URL is not set. For the allotted project requirement, configure DATABASE_URL to the MySQL URI shown above.

18. OUTPUT 1 - LOGIN SCREEN

Student/Admin login screen for secure system access.

SafeCampus | Disaster Preparedness Education

Student / Admin Login

Learn, practice and prepare for emergencies.

Disaster Preparedness & Response Education

Earthquake Flood Fire Cyclone

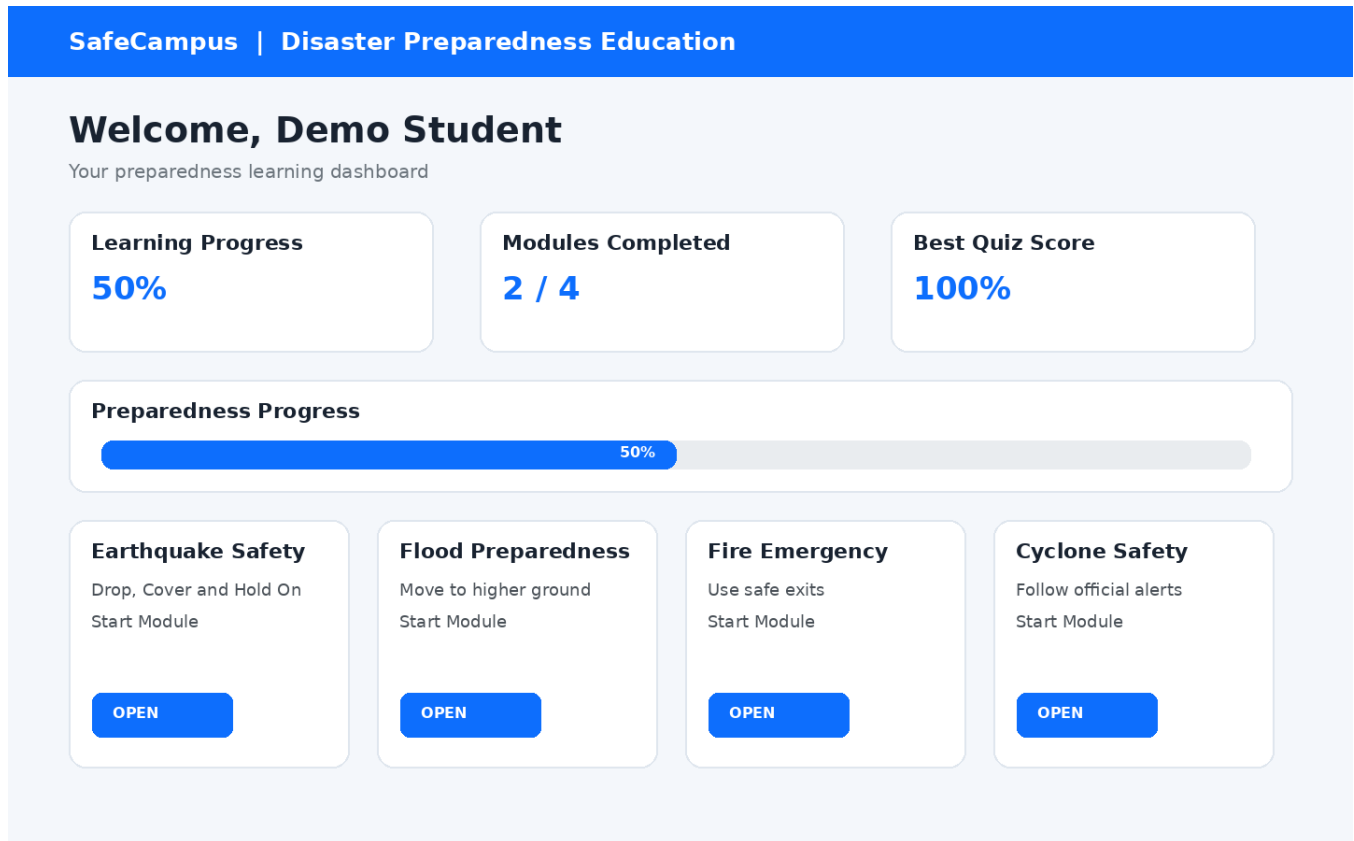
Email

Password

LOGIN SECURELY

19. OUTPUT 2 - STUDENT DASHBOARD

Displays learning progress, completed modules, score and points.



20. OUTPUT 3 - DISASTER LEARNING MODULES

Shows earthquake, flood, fire and cyclone learning content.

SafeCampus | Disaster Preparedness Education

Disaster Learning Modules

Choose a module and learn what to do before, during and after an emergency.

Earthquake Safety

Before: secure shelves and know exits.
During: Drop, Cover and Hold On.
After: expect aftershocks.

Flood Preparedness

Watch official warnings.
Move to higher ground early.
Avoid moving floodwater.

Fire Emergency

Raise alarm.
Use nearest safe staircase.
Never use lifts.

Cyclone & Severe Weather

Prepare emergency supplies.
Stay away from windows.
Follow evacuation instructions.

Disaster Preparedness and Response Education System - CO5_AT3

21. OUTPUT 4 - INTERACTIVE QUIZ

Students answer safety questions and receive a score/badge.

SafeCampus | Disaster Preparedness Education

Interactive Preparedness Quiz

Test your emergency-response knowledge.

1. During an earthquake indoors, the safest first action is?

- ☐ Run outside immediately
- ☒ Drop, Cover and Hold On
- ☐ Use the lift
- ☐ Stand near a window

2. During a flood, which action is safest?

- ☐ Walk through moving water
- ☒ Move to higher ground
- ☐ Drive through flooded roads

Score: 100% | Badge Earned

22. OUTPUT 5 - MOCK EMERGENCY SCENARIO

Simulated fire scenario helps students practice safe decision-making.

SafeCampus | Disaster Preparedness Education

Mock Emergency Scenario

Practice a safe response without real-world danger.

Scenario: Fire alarm during class

Smoke is visible near one corridor. You are on the second floor.

Choose the safest response:

Use the lift to leave quickly

Raise alarm, use safe staircase and assemble

Return to collect your bag

Excellent response: proceed calmly to the designated assembly point.

23. OUTPUT 6 - TEACHER / ADMIN DASHBOARD

Teacher/admin can monitor participation, scores and preparedness status.



24. OUTPUT 7 - CERTIFICATE

Preparedness certificate output for successful learning participation.

SafeCampus | Disaster Preparedness Education

Certificate Output



25. EXPECTED RESULT

The developed system provides continuous and interactive disaster-safety education for schools and colleges. Students can study safety procedures, complete quizzes, practice mock scenarios, track their progress, earn points and view certificates. Teachers and administrators can use the dashboard to identify learning progress and preparedness gaps.

26. CONCLUSION

The Disaster Preparedness and Response Education System converts disaster-awareness training into a structured digital learning platform. The implementation directly supports learning modules, interactive quizzes, mock scenarios, progress tracking, scores, certificates and teacher/admin monitoring using the technologies specified in the allotted problem.